

Context-Window Degradation and Its Impact on Long-Document Research Synthesis

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Large language models (LLMs) increasingly support context windows that are large enough to hold books, reports, codebases, or collections of research papers. A large nominal context window, however, does not guarantee that every part of that context will be used equally well. This review examines *context-window degradation*: the decline in a model’s ability to locate, weight, and integrate information as context length and distractor density increase. The problem is particularly important for long-document research synthesis, where evidence is distributed across many sources and the model must compare, reconcile, and aggregate claims rather than retrieve a single fact. Drawing on studies of positional bias, length generalization, long-context benchmarks, and retrieval-augmented generation (RAG), this paper organizes the evidence around four interacting mechanisms: primacy–recency bias, positional-encoding extrapolation failure, attention dilution, and distractor accumulation. It then compares mitigation families including RAG, hybrid routing, long retrieval units, attention-sink management, positional-encoding extensions, and context compression. The review also highlights an important methodological issue: much of the current evidence comes from retrieval or synthetic benchmarks, while research synthesis requires source attribution, cross-document comparison, evidence balancing, and faithful aggregation. The paper therefore identifies synthesis-specific benchmarking, faithfulness metrics, position-invariant evaluation, and adaptive hybrid architectures as important directions for future work. The discussion is intentionally structured in a way that connects technical mechanisms to practical research-synthesis workflows.

Additional Key Words and Phrases: long-context language models, context-window degradation, lost in the middle, positional bias, retrieval-augmented generation, research synthesis, long-context benchmarks, attention

1 Introduction

The Transformer architecture changed sequence modeling by replacing recurrent processing with self-attention, allowing a model to relate tokens across a sequence in parallel [14]. Since then, one of the most visible directions in large language model (LLM) development has been the expansion of the context window. Early systems commonly operated with a few thousand tokens, while newer systems advertise context lengths ranging from tens of thousands to hundreds of thousands or more. The attraction is straightforward: if an entire report, book, codebase, or collection of scientific papers can be placed inside one prompt, the model appears to need less external retrieval and less manual document chunking.

For research synthesis, this promise is especially appealing. A literature review does not normally ask only for one fact. Instead, it may require a system to identify claims from several papers, compare methodologies, determine where results agree or disagree, track limitations, and produce a balanced synthesis with appropriate attribution. In such a setting, the important information can occur anywhere in the input and may need to be combined with information located far away from it.

Empirical work shows that this assumption of uniform context use is unsafe. Liu et al. [11] found that language models can perform substantially better when relevant information appears near the beginning or end of a long context than when the same information is placed in the middle. This pattern, commonly called *lost in the middle*, indicates that nominal context capacity and effective context utilization are not necessarily the same thing. RULER further showed that performance can fall as context length and task complexity increase even when a model performs nearly perfectly on simpler needle-in-a-haystack retrieval tests [2]. LongBench similarly demonstrated that long-context understanding remains challenging across realistic tasks such as single-document question answering, multi-document question answering, summarization, and few-shot learning [1].

These findings matter because research synthesis is structurally different from single-fact retrieval. If a model overlooks one relevant paragraph, the result may be an incorrect answer. If it systematically underweights several sources in the middle of a long input, the result can be more subtle: the generated text may be fluent and locally correct while still giving a distorted representation of the literature. The issue is therefore not only *whether* a model can retrieve a fact, but whether it can maintain an appropriate balance of evidence across a large set of sources.

This review studies that problem through four questions:

- (1) What mechanisms cause effective context use to degrade as input length grows?
- (2) What evidence demonstrates these effects beyond simple needle-in-a-haystack retrieval?
- (3) How do long-context processing, RAG, and hybrid approaches differ in their strengths and limitations?
- (4) What research gaps remain specifically for long-document research synthesis?

The rest of the paper is organized as follows. Section 2 describes the scope and organization of the review. Section 3 introduces self-attention, positional encoding, and the distinction between nominal and effective context. Section 4 reviews the main empirical evidence. Section 5 explains the mechanisms behind degradation. Section 6 compares mitigation strategies. Section 7 connects these findings to research-synthesis workflows. Section 9 identifies research gaps and future directions, and Section 10 concludes the review.

2 Review Scope and Organization

This paper is a structured narrative review rather than a claim of exhaustive systematic coverage. The literature was organized around four evidence groups: (1) foundational Transformer and attention mechanisms, (2) positional encoding and length generalization, (3) empirical long-context evaluation, and (4) architectural or pipeline-level mitigation strategies. The review prioritizes studies that directly examine long-context behavior or provide evidence relevant to multi-document reasoning and synthesis.

The review emphasizes the central observation that context-window degradation is multi-causal and especially relevant to long-document research synthesis. The selected literature spans foundational work from 2017 onward, with particular attention to empirical studies published between 2023 and 2025.

The review is organized to connect the technical causes of context-window degradation with the practical requirements of research synthesis. Rather than treating long-context performance as a single capability, it separates the problem into positional effects, length generalization, attention dilution, and distractor accumulation. It then compares evaluation benchmarks and mitigation strategies before considering what these findings mean for evidence coverage, attribution, and balanced synthesis. Three themes guide the discussion: the difference between nominal and effective context, the distinction between retrieval and aggregation, and the need to evaluate synthesis quality rather than retrieval accuracy alone.

3 Background: Why Long Context Is Difficult

3.1 Self-Attention and the Scaling Problem

The Transformer is built around self-attention, which allows each token to interact with representations at other positions [14]. The scaled dot-product attention operation is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (1)$$

where Q , K , and V are query, key, and value matrices and d_k is the key dimension. The product $QK^\top$ produces pairwise interactions between input positions. For a sequence of length n , this creates an $n \times n$ attention-score matrix, giving the standard full-attention computation a quadratic dependence on sequence length.

Equation 1 explains an important distinction. Increasing the maximum context length is not simply a matter of storing more tokens. A model must both process more pairwise interactions and learn how to interpret positional relationships over a larger range. Surveys of long-context Transformer architectures therefore distinguish among changes to pretraining, attention structure, positional representations, memory, and inference pipelines [3, 10].

3.2 Positional Information and Length Generalization

Self-attention by itself does not encode the order of tokens. Positional information is therefore introduced through mechanisms such as absolute position embeddings, relative position representations, Rotary Position Embedding (RoPE), and Attention with Linear Biases (ALiBi). RoPE incorporates positional information through rotations in the attention representation and has become a widely used design for Transformer language models [13]. ALiBi instead adds a distance-dependent bias to attention scores and was designed to improve extrapolation to sequences longer than those seen during training [12].

However, the choice of positional representation does not remove the generalization problem. Kazemnejad et al. [6] systematically compared several positional-encoding schemes and found that common explicit methods, including ALiBi and Rotary, can still struggle with length generalization on downstream reasoning tasks. This suggests that context extension involves both architectural design and learned behavior.

3.3 Nominal Context Versus Effective Context

A model may accept a long sequence at the interface level without using every part of that sequence equally well. It is therefore useful to distinguish between *nominal context length*, the maximum number of tokens a system accepts, and *effective context*, the range over which the model can reliably use information for a particular task.

RULER makes this distinction especially clear. It evaluates several long-context tasks at increasing sequence lengths and reports substantial performance degradation even for models that perform very well on simple retrieval tests [2]. Importantly, effective context is task-dependent: a model can retain useful performance for one task while degrading much earlier for another task that requires multi-hop reasoning or aggregation.

4 Empirical Evidence of Context Degradation

4.1 From Needle-in-a-Haystack to Richer Evaluation

The needle-in-a-haystack (NIAH) setup places a small piece of relevant information inside a long distractor context and measures whether the model can retrieve it. This diagnostic is intuitive and useful for testing basic recall, but it is limited because it can reduce long-context understanding to finding one inserted fact. Kamradt’s widely used implementation helped popularize this style of stress test [5].

RULER extends this idea by varying the number and type of needles and adding multi-hop tracing and aggregation tasks [2]. LongBench takes a complementary approach by evaluating realistic tasks across 21 datasets and six task categories, including single-document and multi-document question answering, summarization, few-shot learning, synthetic tasks, and code completion [1].

Table 1. Comparison of representative long-context evaluation approaches.

Benchmark approach	Evaluation type	What it tests	Main contribution	Limitation for research synthesis
NIAH [5]	Synthetic retrieval	Finding a single inserted fact at controlled positions	Simple diagnostic for positional recall	Does not require evidence balancing, comparison, or multi-source synthesis
RULER [2]	Synthetic, configurable	Multiple needles, multi-hop tracing, aggregation	Shows degradation beyond single-fact retrieval and separates length from task complexity more explicitly	Tasks remain synthetic and do not fully reproduce open-ended literature review
LongBench [1]	Realistic multi-task benchmark	QA, summarization, few-shot learning, code, single- and multi-document tasks	Broader and more realistic evaluation of long-context understanding	Automatic task metrics do not directly measure attribution or balance of evidence
LaRA [8]	Comparative LC vs. RAG evaluation	Practical QA over naturally occurring long texts	Shows that architecture choice depends on model, task, context length, and retrieval quality	Focuses primarily on QA rather than open-ended synthesis

These benchmarks therefore move the discussion from “Can the model find one item?” toward “Can the model use a long context to perform a meaningful task?”

Table 1 compares the main evaluation approaches discussed in this review.

4.2 Lost in the Middle

Liu et al. [11] provide the clearest evidence of positional sensitivity. They evaluated multi-document question answering and key-value retrieval while systematically moving relevant information through the context. Performance was generally strongest when the relevant information appeared near the beginning or end and weaker when it appeared in the middle. The important point is not merely the shape of the curve; it is that the same information can receive different treatment depending on where it is placed.

Figure 1 illustrates this idea schematically. The figure is conceptual rather than a reproduction of a particular model’s numerical results. It is included to clarify the relationship reported across the literature: position can act as an unintended source of importance.

For research synthesis, the implication is more serious than a retrieval error. Suppose ten research papers are concatenated in arbitrary order. If papers five through eight contain important evidence, their location alone should not reduce their influence. Yet a position-sensitive model may underuse those sources. The ordering of the input can therefore become an unintended variable in the synthesis process.

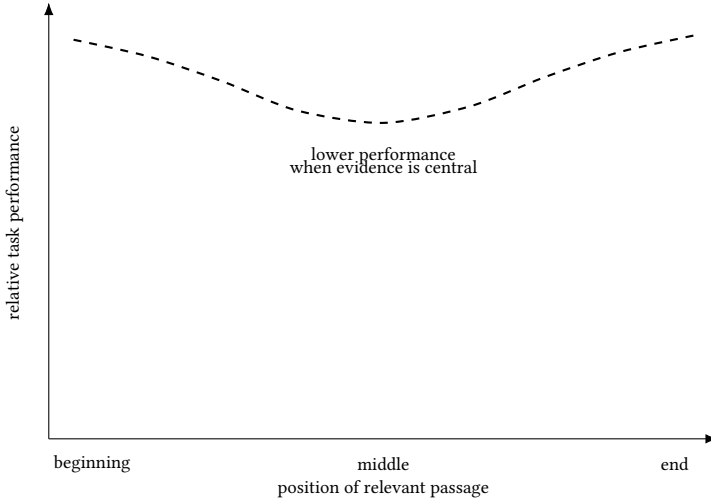


Fig. 1. Conceptual representation of the positional sensitivity reported in long-context evaluations. The same relevant information can be easier to use when placed near the beginning or end than when placed in the middle.

4.3 Aggregation Is Harder Than Retrieval

RULER reports that aggregation and multi-hop tasks are more revealing than simple retrieval as context length increases [2]. This is directly relevant to literature review. A synthesis question may require a model to combine several partial observations, such as identifying which papers use a particular dataset, determining which studies report a similar limitation, and then explaining why their findings differ. No single passage contains the answer; the answer emerges only after aggregation.

This distinction can be represented as follows. Single-location retrieval asks the model to map a query to one relevant span. Research synthesis asks it to construct a supported claim from multiple spans:

$$\text{Synthesis quality} \approx f(\text{coverage, integration, attribution, faithfulness}), \quad (2)$$

where the expression is a conceptual decomposition used in this review, not a standardized benchmark metric. The four terms highlight why exact-match or retrieval accuracy alone is insufficient for evaluating research synthesis.

5 Mechanisms of Context-Window Degradation

The evidence suggests that context-window degradation is not one isolated failure mode. Instead, several mechanisms interact. Figure 2 summarizes the relationship between the mechanisms and the effective context used by the model.

5.1 Primacy–Recency Bias

The lost-in-the-middle effect is a positional bias in which information near the beginning and end of a context can be used more effectively than information in the middle [11]. This can arise even when the information itself is unchanged. For synthesis systems, it means that document ordering is potentially consequential. A paper should matter because of the evidence it contains, not because it happened to be inserted early in a prompt.

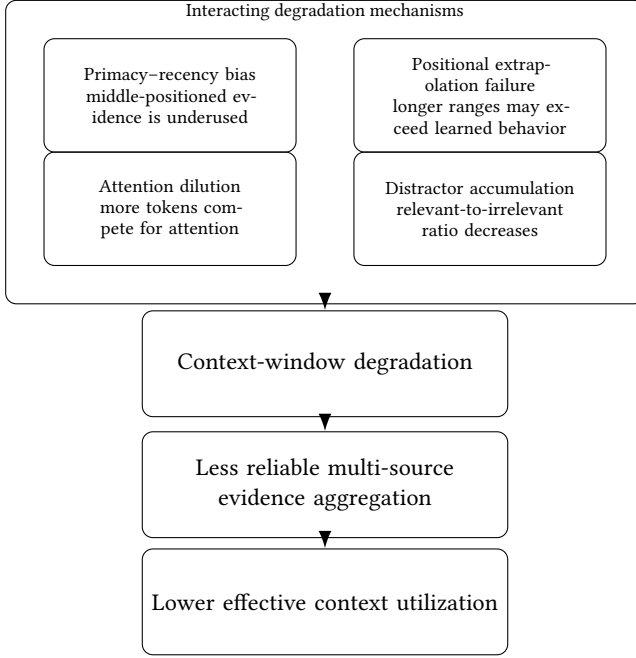


Fig. 2. Four interacting mechanisms contribute to context-window degradation. The grouped representation avoids implying a strict causal order among the mechanisms; together, they can reduce the reliability of multi-source evidence aggregation and effective context utilization.

5.2 Positional-Encoding Extrapolation Failure

A second mechanism occurs when inference length extends beyond the range or distribution for which the model learned reliable positional behavior. RoPE and ALiBi were designed to provide useful positional inductive biases, but systematic experiments show that explicit positional schemes do not automatically guarantee robust length generalization [6]. Long-context surveys consequently treat position representation and context extension as active design problems rather than solved components [3, 10].

5.3 Attention Dilution and Aggregation Failure

As the number of tokens increases, the model must process more candidate interactions. In a long input, many tokens may be weakly relevant or unrelated to the current task. This can make it harder for the model to preserve a strong signal for every useful passage. The attention-sink phenomenon provides a related perspective: Xiao et al. [15] showed that some initial tokens receive unusually important attention scores and that preserving these sink tokens can stabilize generation over very long streams.

For research synthesis, the deeper issue is aggregation. If ten relevant claims are dispersed across a context, the model must maintain enough information about all ten and combine them. A system that is adequate at retrieving one fact can still fail at counting, comparing, or reconciling multiple facts. RULER’s multi-hop and aggregation tasks provide empirical support for this distinction [2].

Table 2. Comparison of major mitigation strategies for context-window degradation.

Approach	Primary target	Reported benefit	Research-synthesis value	Key limitation
RAG [7]	Distractors and excessive input length	Retrieves a smaller set of relevant passages before generation	Improves relevance density and can provide explicit source passages	Retriever can miss evidence; fragmented chunks may lose cross-document context
Long-context processing	Avoids aggressive retrieval/chunking	Preserves access to the original document and supports broad comparison	Useful when evidence is distributed across many locations	Higher compute/token cost and persistent long-context degradation
Hybrid routing / Self-Route [9]	Task-dependent choice between RAG and LC	Can reduce cost while maintaining performance close to a strong LC baseline in the studied settings	Allows simple queries to use retrieval while harder synthesis uses broader context	Depends on reliable routing/self-assessment; synthesis-specific validation is limited
Long retrieval units [4]	Context loss from small chunks	Uses larger retrieval units to preserve local context and reduce retriever burden	Better preserves surrounding evidence before synthesis	Larger retrieved units can reintroduce long-context noise
Attention-sink management [15]	Long-stream instability and attention allocation	Supports stable generation beyond the trained window in streaming settings	Relevant to very long streams and memory-efficient inference	Designed primarily for streaming stability, not evidence-balanced synthesis
Positional extensions [6, 12, 13]	Position generalization	Improve or extend usable sequence length under particular settings	Can reduce failure caused by position range	Longer nominal length does not guarantee uniform evidence use
Context compression / curation	Length and distractor density	Reduces token count and irrelevant material before generation	Can improve relevance density and cost	Compression may delete evidence or introduce factual drift and attribution loss

5.4 Distractor Accumulation

Long source documents contain substantial material that is not directly relevant to the synthesis question: background sections, implementation details, repeated explanations, references, and unrelated findings. As context grows, the ratio of useful to irrelevant material can decrease. This is important because simply increasing the context window does not guarantee that the model will identify the correct evidence among all available text.

This mechanism also explains why retrieval and context compression remain relevant even when long-context models are available. A shorter context can be advantageous when it increases the density of task-relevant information, although aggressive compression introduces a risk of deleting evidence or weakening source attribution.

6 Current Mitigation Approaches

Mitigation strategies can be grouped into pipeline-level methods, hybrid routing, retrieval-unit re-design, and model-level changes. Table 2 compares these families using the degradation mechanism they primarily address, their main benefit, and their limitation for research synthesis.

6.1 Retrieval-Augmented Generation

RAG combines a language model with an external retrieval mechanism. Instead of placing an entire corpus into the prompt, a retriever selects passages that are likely to be useful and the generator conditions on those passages [7]. This reduces the number of tokens the generator must process and can improve the density of relevant information.

However, RAG introduces a new failure mode: retrieval error. If a relevant source is not retrieved, the generator cannot use it. This is especially important for research synthesis because a paper that

is only moderately similar to the query may still contain a critical counterexample or limitation. Retrieval therefore trades one problem for another: it reduces distractor load but can reduce coverage.

6.2 Long Context Versus RAG

Comparative studies do not support a single universal winner. Li et al. [9] found that, with sufficient resources, long-context processing could achieve higher average performance across their evaluated datasets, while RAG offered a substantial cost advantage. Their Self-Route approach dynamically selected between RAG and long-context processing. Xu et al. [16] similarly found that a smaller-context model with retrieval could achieve performance comparable to a larger-context setup on some long-context tasks at lower computational cost.

More recent evaluation with LaRA reports that the preferable choice depends on factors including model capability, task type, context length, and the characteristics of retrieved chunks [8]. Taken together, these studies suggest that “RAG versus long context” is not a binary question. The useful choice depends on what the task needs: broad coverage, precise retrieval, cross-document reasoning, cost efficiency, or some combination.

6.3 Long Retrieval Units and Hybrid Designs

One limitation of conventional RAG is that small chunks can remove information that gives a retrieved passage its meaning. Jiang et al. [4] proposed LongRAG, which uses substantially larger retrieval units and then relies on a long-context reader. The idea shifts some burden from retrieval to the reader: retrieval identifies a broader region, while the reader extracts the exact evidence from that region.

This approach is conceptually relevant to research synthesis because it addresses an important tension. Very small chunks improve retrieval precision but can break the relationships between claims, methods, and limitations. Very large chunks preserve context but increase the amount of material the model must process. Long retrieval units attempt to find a middle ground rather than choosing one extreme.

6.4 Attention and Positional Interventions

StreamingLLM demonstrates that managing attention structure can improve stability for inputs that continue beyond the original training window [15]. Positional methods such as ALiBi and RoPE provide different mechanisms for representing position and supporting longer sequences [12, 13]. These methods address an architectural side of the problem, whereas RAG and compression address the input pipeline.

The distinction is useful: architecture-level methods attempt to make the model itself better at long sequences, while pipeline-level methods attempt to make the sequence easier to use. A practical research-synthesis system may need both.

6.5 Prompt Compression and Context Curation

Compression can be extractive, abstractive, or learned. Its basic goal is to reduce the number of tokens while preserving the information needed for the task. This can simultaneously reduce sequence length and distractor density.

The main risk is information loss before the final synthesis stage. If a compressor removes a small qualification such as “under these experimental conditions,” a later summary may turn a conditional result into a general claim. For research writing, source attribution is also important: compression should preserve enough provenance to identify which source supports which statement.

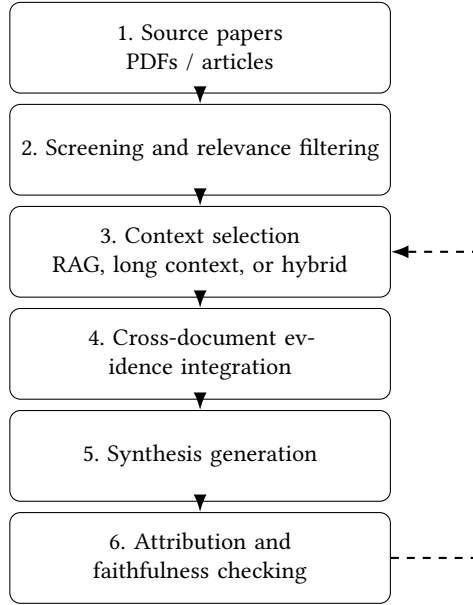


Fig. 3. Conceptual workflow for long-document research synthesis. The dashed feedback path indicates that verification can trigger revision of context selection when evidence coverage or attribution is inadequate.

7 Implications for Long-Document Research Synthesis

The central argument of this review is that long-document research synthesis should be treated as a distinct workload rather than as a larger version of question answering. Figure 3 presents a practical conceptual pipeline.

7.1 Coverage Versus Relevance

A long-context system can offer high coverage because many documents can be provided at once. RAG can offer high relevance density because it filters the corpus before generation. For research synthesis, neither property is sufficient by itself. High coverage without relevance can produce distractor overload; high relevance without coverage can omit important counterevidence.

This creates a practical design question: how much of the corpus should be shown to the model before synthesis? The answer is task-dependent. A question about one specific result may benefit from narrow retrieval, whereas a review question about trends across an entire research area may require broader coverage and explicit aggregation.

7.2 Attribution and Evidence Balance

A synthesis should not merely mention sources; it should preserve the relationship between claims and evidence. This is where long-context degradation can be particularly subtle. Suppose a model retrieves the strongest positive result from one paper but underuses a limitation described in another paper located in the middle of the context. The final paragraph can remain factually plausible while becoming unbalanced.

This motivates evaluation dimensions beyond answer accuracy. A research-synthesis evaluation should ask at least four questions: (1) Were the relevant sources covered? (2) Were their claims interpreted correctly? (3) Were conflicting or qualifying findings preserved? (4) Can a reader trace

important claims back to their sources? The conceptual decomposition in Equation 2 reflects these requirements.

7.3 Why Ordering Should Be Treated as an Experimental Variable

If positional effects exist, document order can influence results. Therefore, an evaluation that always places the most relevant source first may overestimate real-world performance. A stronger protocol would vary source order while keeping the evidence set fixed. If the synthesis changes substantially under permutation, the system is sensitive to input ordering.

This suggests a simple position-invariance test for future research: construct multiple permutations of the same source set, run the synthesis under identical instructions, and measure how much the resulting evidence coverage and claims change. Such a protocol would directly test whether a model treats evidence according to content rather than location.

8 Research Challenges and Limitations of Current Evidence

8.1 Benchmark–Task Mismatch

A large portion of the evidence base comes from retrieval, question answering, or synthetic tasks. RULER improves on simple NIAH by adding aggregation and multi-hop tasks, while LongBench includes realistic summarization and multi-document tasks [1, 2]. Nevertheless, neither benchmark fully captures the open-ended character of a research review, where the system must decide which evidence matters, reconcile conflicting claims, and communicate uncertainty.

8.2 Faithfulness Is Not the Same as Accuracy

Traditional accuracy metrics are often based on whether a specific answer is correct. Research synthesis has a different requirement. A generated paragraph can contain individually correct statements while still misrepresenting the overall literature. For example, selectively emphasizing positive findings can produce a misleading summary even if every sentence is technically true.

This motivates faithfulness-specific measures that evaluate source attribution, evidence coverage, contradiction handling, and whether the relative strength of claims is preserved. Such metrics are currently less standardized than retrieval or answer-level accuracy.

8.3 Length and Complexity Are Confounded

As contexts become longer, they often become more difficult for reasons other than token count. Longer documents may contain more topics, more distractors, more repeated information, and more contradictions. It is therefore difficult to determine whether a performance drop is caused by sequence length itself or by the increase in semantic complexity.

RULER’s configurable synthetic design helps isolate some variables, but its authors also note that synthetic tasks cannot replace realistic evaluation [2]. A useful future benchmark should vary length and complexity independently wherever possible.

8.4 Rapid Model Turnover

Long-context capabilities are changing quickly. Results measured on one generation of models may not describe later systems. Recent benchmark evidence already shows that some contemporary models can perform strongly at lengths where earlier models degraded, while more complex aggregation tasks can remain difficult [2]. This means that the research question should focus not only on whether a particular model has a lost-in-the-middle curve, but also on which mechanisms persist across model generations.

9 Research Gaps and Future Directions

The preceding sections point to several research directions that are specifically relevant to research synthesis.

9.1 Synthesis-Specific Benchmarks

A useful benchmark should contain sets of related research papers rather than isolated documents and should require the model to produce a structured synthesis. Tasks could include: identifying consensus, identifying disagreement, comparing methods, tracing limitations, and determining whether a conclusion is supported by multiple independent sources. Evaluation should include both automatic and human-judged components.

9.2 Position-Invariant Evaluation

Long-context evaluation should vary the order of documents and relevant passages. If the same evidence set produces different conclusions under different permutations, the benchmark can quantify positional sensitivity directly. This would complement the existing lost-in-the-middle diagnostic by moving from a single inserted fact to a complete evidence set.

9.3 Faithfulness and Attribution Metrics

Future work should measure at least four properties separately: evidence coverage, claim correctness, attribution correctness, and balance across sources. A model should not receive full credit merely because it produces a fluent answer. The evaluation should penalize unsupported claims, missing counterevidence, and incorrect source attribution.

9.4 Adaptive Hybrid Architectures

The evidence reviewed in Section 6 suggests that no single processing strategy is universally appropriate. Hybrid routing is therefore a promising direction. A system could first estimate the synthesis task’s requirements and then decide whether to use narrow retrieval, broad long-context processing, or a combination. Li et al. [9] provide an early example of this idea, while LaRA shows why the routing decision is sensitive to task and model characteristics [8].

For research synthesis, a more advanced router could consider not only query difficulty but also evidence distribution. If relevant evidence is concentrated in a few documents, retrieval may be sufficient. If evidence is distributed across many sources and requires comparison, broader context may be useful. This hypothesis needs dedicated empirical testing.

9.5 Evidence-Preserving Compression

Compression should be designed around evidence rather than token count alone. A research-oriented compressor should preserve claims, qualifications, numerical findings, methodological conditions, and source identity. This could reduce context length without turning a careful literature review into a sequence of disconnected summaries.

9.6 Separating Model Capability From Pipeline Quality

Finally, evaluations should report the performance of the underlying long-context model separately from the performance of retrieval, reranking, compression, and routing components. Otherwise, a poor retrieval result may be interpreted as a model limitation, or a strong curated context may hide weaknesses in the model’s ability to locate evidence independently.

10 Conclusion

Context-window expansion has made it possible to place increasingly large collections of text into a single LLM prompt, but nominal capacity should not be treated as equivalent to reliable context use. The literature reviewed in this paper shows that long-context behavior is affected by positional bias, positional-encoding generalization, attention dilution, and distractor accumulation. These mechanisms interact, making tasks that require aggregation across many dispersed pieces of evidence especially challenging.

For long-document research synthesis, the problem is more consequential than simple retrieval failure. A system can produce fluent and locally accurate text while still underrepresenting sources, missing counterevidence, or weakening attribution. RAG, hybrid routing, long retrieval units, attention management, positional extensions, and context compression each address different parts of the problem, but comparative studies show that their usefulness depends on the task, model, context length, retrieval quality, and computational budget.

The most important implication is therefore methodological: long-context systems should be evaluated on the kind of work we actually expect them to perform. For research synthesis, this means testing multi-source aggregation, evidence balance, source attribution, and robustness to document ordering. Future systems will likely benefit from adaptive pipelines that combine retrieval and long-context processing rather than assuming that a larger context window alone solves the problem. In this sense, the central research question is shifting from “How many tokens can the model accept?” to “How reliably can the model use the evidence contained in those tokens?”

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